Database System Engineering and Distributed Backend Development

Course Code: 25CS1302E

Sql-commands-assignment

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Section - 3

Create the database:

```
CREATE DATABASE IF NOT EXISTS bookflow3_db;
```

Output:

```
35  13:38:09  CREATE DATABASE IF NOT EXISTS bookflow3_db
```

Select the database:

```
USE bookflow3_db;
```

Output:

```
37  13:40:31  USE bookflow3_db
```

Create Table:

```
CREATE TABLE student_marks1 (
    roll_no INT PRIMARY KEY,
    name VARCHAR(50),
    subject VARCHAR(50),
    marks DECIMAL(5,2)
);
```

Output:

```
38  13:42:10  CREATE TABLE student_marks1 (    roll_no INT PRIMARY KEY,
```

```
SELECT * FROM student_marks1;
```

	roll_no	name	subject	marks
*	NULL	NULL	NULL	NULL

Insert Data:

```
INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES
(1, 'Likhitha', 'Math', 85.50),
(2, 'Akhila', 'Math', 92.75),
(3, 'Shanmukh', 'Math', 78.40),
(4, 'Vaishnavi', 'Math', 88.90),
(5, 'Aswitha', 'Math', 80.25),
(6, 'Sushma', 'aws', 98.50),
(7, 'Sowmya', 'dbms', 95.75),
(8, 'Mokshagna', 'english', 97.40),
(9, 'Rani', 'aws1', 99.90),
(10, 'Vignesh', 'azure', 82.25);
```

Output:

```
41  13:52:01  INSERT INTO student_marks1 (roll_no, name, subject, marks) VALUES (1, 'Likhitha', 'Math', 
```

Simple Select:

```
SELECT * FROM student_marks1;
```

Output:

roll_no	name	subject	marks
1	Likhitha	Math	85.50
2	Akhila	Math	92.75
3	Shanmukh	Math	78.40
4	Vaishnavi	Math	88.90
5	Aswitha	Math	80.25
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
8	Mokshagna	english	97.40
9	Rani	aws1	99.90
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

<u>Aggregate Functions:</u>

1) `SELECT COUNT(*) AS total_students FROM student_marks1;`

Output:

total_students
10

2) `SELECT SUM(marks) FROM student_marks1;`

Output:

	SUM(marks)
▶	899.60

3)
```
SELECT AVG(marks) FROM student_marks1;
```

Output:

	AVG(marks)
▶	89.960000

4)
```
SELECT MAX(marks) FROM student_marks1;
```

Output:

	MAX(marks)
▶	99.90

5)
```
SELECT MIN(marks) FROM student_marks1;
```

Output:

	MIN(marks)
▶	78.40

Filtering with WHERE

1)
```
SELECT * FROM student_marks1 WHERE marks > 85;
```

Output:

	roll_no	name	subject	marks
▶	1	Likhitha	Math	85.50
	2	Akhila	Math	92.75
	4	Vaishnavi	Math	88.90
	6	Sushma	aws	98.50
	7	Sowmya	dbms	95.75
	8	Mokshagna	english	97.40
	9	Rani	aws1	99.90
	NULL	NULL	NULL	NULL

2)
```
SELECT * FROM student_marks1 WHERE marks >= 90;
```

Output:

roll_no	name	subject	marks
2	Akhila	Math	92.75
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
8	Mokshagna	english	97.40
9	Rani	aws1	99.90
NULL	NULL	NULL	NULL

3)
```
SELECT * FROM student_marks1 WHERE marks < 80;
```

Output:

roll_no	name	subject	marks
3	Shanmukh	Math	78.40
NULL	NULL	NULL	NULL

4)
```
SELECT * FROM student_marks1 WHERE marks BETWEEN 80 AND 90;
```

Output:

roll_no	name	subject	marks
1	Likhitha	Math	85.50
4	Vaishnavi	Math	88.90
5	Aswitha	Math	80.25
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

5)
```
SELECT * FROM student_marks1 WHERE name LIKE 'S%';
```

Output:

roll_no	name	subject	marks
3	Shanmukh	Math	78.40
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
NULL	NULL	NULL	NULL

6)
```
SELECT * FROM student_marks1 WHERE name IN ('Vignesh', 'Likhitha', 'Rani');
```

Output:

roll_no	name	subject	marks
1	Likhitha	Math	85.50
9	Rani	aws1	99.90
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

7)
```
SELECT * FROM student_marks1 WHERE marks > 85 AND (subject = 'Math' OR name LIKE 'S%');
```

Output:

roll_no	name	subject	marks
1	Likhitha	Math	85.50
2	Akhila	Math	92.75
4	Vaishnavi	Math	88.90
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
NULL	NULL	NULL	NULL

Update Queries

1)
```
UPDATE student_marks1 SET marks = 90.00 WHERE roll_no = 3;
SELECT * FROM student_marks1;
```

Output:

roll_no	name	subject	marks
1	Likhitha	Math	85.50
2	Akhila	Math	92.75
3	Shanmukh	Math	90.00
4	Vaishnavi	Math	88.90
5	Aswitha	Math	80.25
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
8	Mokshagna	english	97.40
9	Rani	aws1	99.90
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

2)
```
UPDATE student_marks1 SET subject = 'Remedial Math' WHERE marks < 80;
SELECT * FROM student_marks1;
```

Output:

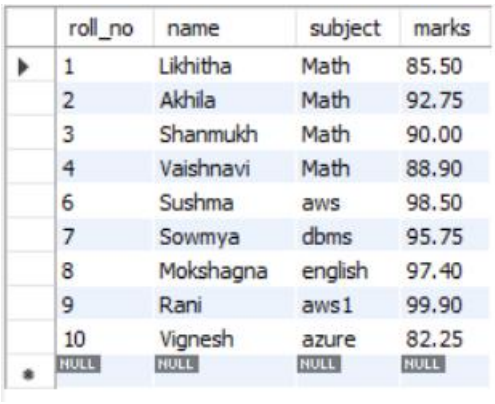
roll_no	name	subject	marks
1	Likhitha	Math	85.50
2	Akhila	Math	92.75
3	Shanmukh	Math	90.00
4	Vaishnavi	Math	88.90
5	Aswitha	Math	80.25
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
8	Mokshagna	english	97.40
9	Rani	aws1	99.90
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

Delete Queries

```
DELETE FROM student_marks1 WHERE roll_no = 5;
```

1)
```
SELECT * FROM student_marks1;
```

Output:

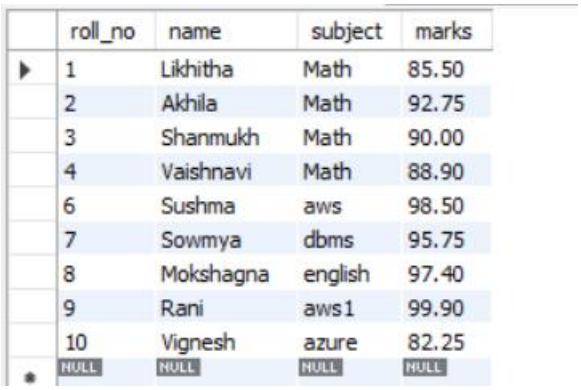


roll_no	name	subject	marks
1	Likhitha	Math	85.50
2	Akhila	Math	92.75
3	Shanmukh	Math	90.00
4	Vaishnavi	Math	88.90
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
8	Mokshagna	english	97.40
9	Rani	aws1	99.90
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

```
DELETE FROM student_marks1 WHERE marks < 75;
```

2)
```
SELECT * FROM student_marks1;
```

Output:

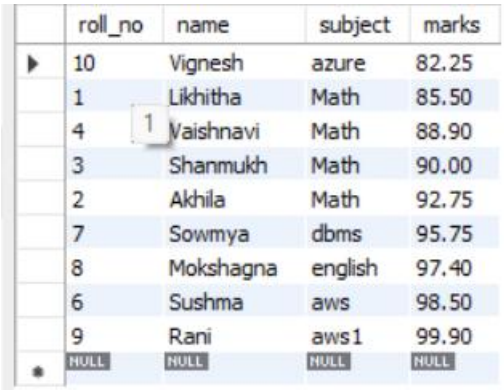


roll_no	name	subject	marks
1	Likhitha	Math	85.50
2	Akhila	Math	92.75
3	Shanmukh	Math	90.00
4	Vaishnavi	Math	88.90
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
8	Mokshagna	english	97.40
9	Rani	aws1	99.90
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

Sorting

1)
```
SELECT * FROM student_marks1 ORDER BY marks ASC;
```

Output:



roll_no	name	subject	marks
10	Vignesh	azure	82.25
1	Likhitha	Math	85.50
4	Vaishnavi	Math	88.90
3	Shanmukh	Math	90.00
2	Akhila	Math	92.75
7	Sowmya	dbms	95.75
8	Mokshagna	english	97.40
6	Sushma	aws	98.50
9	Rani	aws1	99.90
NULL	NULL	NULL	NULL

2)
```
SELECT * FROM student_marks1 ORDER BY marks DESC;
```

Output:

roll_no	name	subject	marks
9	Rani	aws1	99.90
6	Sushma	aws	98.50
8	Mokshagna	english	97.40
7	Sowmya	dbms	95.75
2	Akhila	Math	92.75
3	Shanmukh	Math	90.00
4	Vaishnavi	Math	88.90
1	Likhitha	Math	85.50
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

3)
```
SELECT * FROM student_marks1 ORDER BY name ASC;
```

Output:

roll_no	name	subject	marks
2	Akhila	Math	92.75
1	Likhitha	Math	85.50
8	Mokshagna	english	97.40
9	Rani	aws1	99.90
3	Shanmukh	Math	90.00
7	Sowmya	dbms	95.75
6	Sushma	aws	98.50
4	Vaishnavi	Math	88.90
10	Vignesh	azure	82.25
NULL	NULL	NULL	NULL

4)
```
SELECT * FROM student_marks1 ORDER BY name DESC;
```

Output:

roll_no	name	subject	marks
10	Vignesh	azure	82.25
4	Vaishnavi	Math	88.90
6	Sushma	aws	98.50
7	Sowmya	dbms	95.75
3	Shanmukh	Math	90.00
9	Rani	aws1	99.90
8	Mokshagna	english	97.40
1	Likhitha	Math	85.50
2	Akhila	Math	92.75
NULL	NULL	NULL	NULL

Group By

1)
```
SELECT subject, SUM(marks) AS total_marks FROM student_marks1 GROUP BY subject;
```

Output:

	subject	total_marks
▶	Math	357.15
	aws	98.50
	dbms	95.75
	english	97.40
	aws1	99.90
	azure	82.25

2)
```sql
SELECT subject, AVG(marks) AS avg_marks FROM student_marks1 GROUP BY subject;
```

Output:

	subject	avg_marks
▶	Math	89.287500
	aws	98.500000
	dbms	95.750000
	english	97.400000
	aws1	99.900000
	azure	82.250000

3)
```sql
SELECT subject, COUNT(*) AS num_students FROM student_marks1 GROUP BY subject;
```

Output:

	subject	num_students
▶	Math	4
	aws	1
	dbms	1
	english	1
	aws1	1
	azure	1

Having

1)
```sql
SELECT subject, AVG(marks) AS avg_marks
FROM student_marks1
GROUP BY subject
HAVING AVG(marks) > 90;
```

Output:

	subject	avg_marks
▶	aws	98.500000
	dbms	95.750000
	english	97.400000
	aws1	99.900000

2)
```sql
SELECT subject, COUNT(*) AS num_students
FROM student_marks1
GROUP BY subject
HAVING COUNT(*) > 1;
```

Output:

	subject	num_students
▶	Math	4

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